

MECHANICAL DESIGN PORTFOLIO

NAQEEB ARSHAD

Mechanical Design Engineer

Mechanical design, CAD modelling, CNC manufacturing, fabrication, and product development.

CURRENT ROLE Design Engineer - Tech Valley, Islamabad

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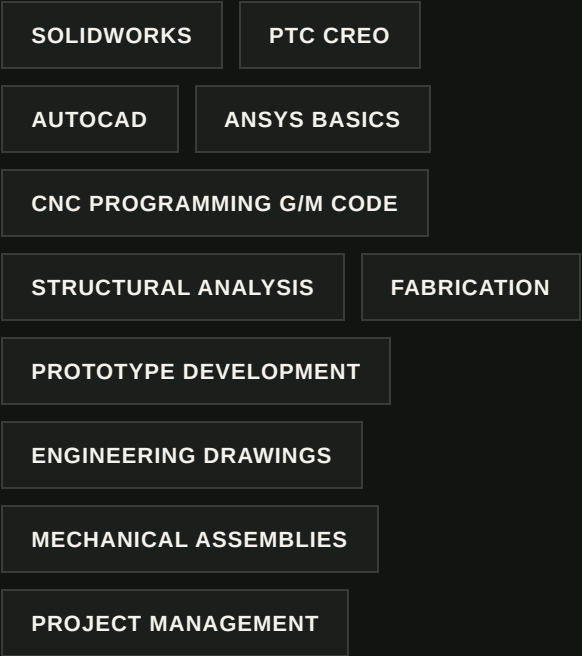
Design. Validate. *Manufacture.*

Mechanical engineering experience across CAD modelling, fabrication, machining, and product development. Precise enough for engineering drawings. Practical enough for manufacturing.

I transform engineering concepts into practical, manufacturable solutions through mechanical design, CAD modelling, engineering drawings, machining preparation, and prototype development.

CURRENT ROLE	Design Engineer - Tech Valley, Islamabad
EDUCATION	BS Mechanical Engineering 3.18 HITEC University Taxila
LOCATION	Islamabad, Pakistan
EMAIL	naqeebarshad402@gmail.com
PHONE	+92 310 5038342

CAPABILITY STACK



Professional Background

Verified roles, education, certifications, and achievements from the supplied portfolio and resume.

PROFESSIONAL EXPERIENCE

Current

Design Engineer

Tech Valley, Islamabad

SolidWorks design of drone-handling, rotary-cleaning, and vehicle-mounted systems.

Jul-Sep 2025

Intern

NUST College of EME (Robot Maker Lab)

Eight-week internship in design, manufacturing, CNC cutting, and fabrication.

Apr-Jun 2022

Assistant Engineer

Electrosolz, Islamabad

Product design and manufacturing work involving 3D printers.

Independent

Freelance CAD Designer

Remote

AutoCAD house plans and SolidWorks mechanical components for client requirements.

EDUCATION

2022-2026

BS Mechanical Engineering

HITEC University Taxila | CGPA 3.18 / 4.00

2019-2021

DAE Mechanical

KRL Institute of Technology Kahuta | 93%

CERTIFICATIONS & ACHIEVEMENTS

+ **NINVAST** - 2nd International Additive Manufacturing and Vacuum Technology Conference

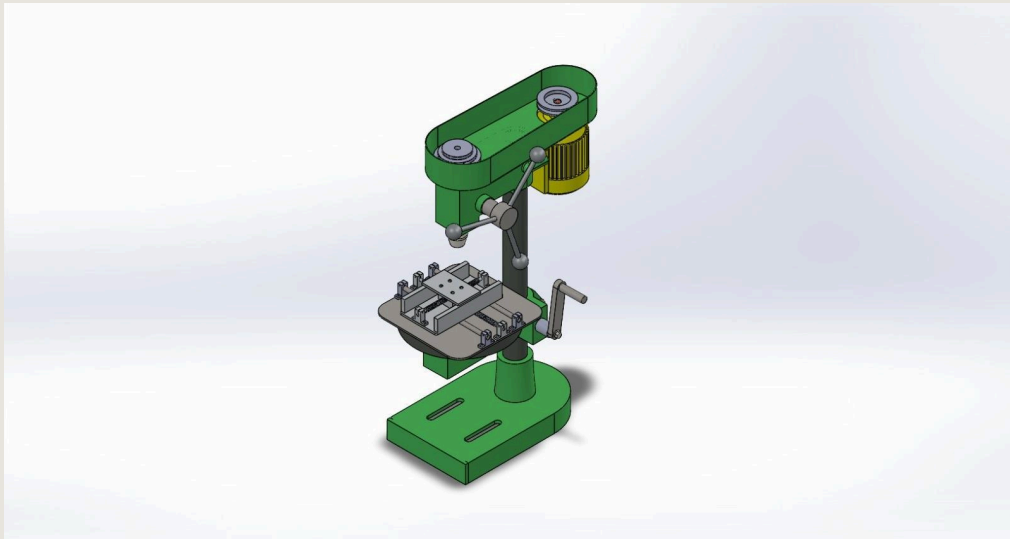
+ **NAVTTTC** - Six-month AutoCAD Diploma (2022)

+ **Technical Head** - Robo Fiesta 8.0 | Two-time HI-Robo Tec paper-plane winner

+ **Letter of Appreciation** - 11th Open House 2025, HITEC University Taxila

Friction-Based Metal Printing Machine (AFSD)

Manual desktop Additive Friction Stir Deposition machine for AA6061 aluminium, completed at HITEC University Taxila.



+46.2%

STRENGTH IMPROVEMENT

+4%

HARDNESS VS. CAST-WELDED AA6061

FINAL YEAR PROJECT

Manual desktop AFSD machine

The complete mechanical design was developed in SolidWorks. Primary load-bearing components were checked through static structural and vibration analysis before fabrication.

- + Complete SolidWorks assembly
- + Static structural analysis
- + Vibration and frequency analysis
- + Successful AFSD deposition in Pakistan

Verified AFSD Results

Results reported in the reference portfolio after design validation, fabrication, deposition, and material testing.

01 / Verified

1st in Pakistan

Successful metal deposition using the developed AFSD process.

02 / Verified

+4% hardness

Improvement over conventionally cast-welded AA6061.

03 / Verified

+46.2% strength

Compression-strength improvement in the deposited specimen.

04 / Verified

24.84 Hz

First natural frequency confirmed by modal analysis.

PROJECT VERIFICATION

01 SolidWorks Design

Complete machine assembly and load-bearing components.

02 Structural Analysis

Static structural checks before fabrication.

03 Vibration Analysis

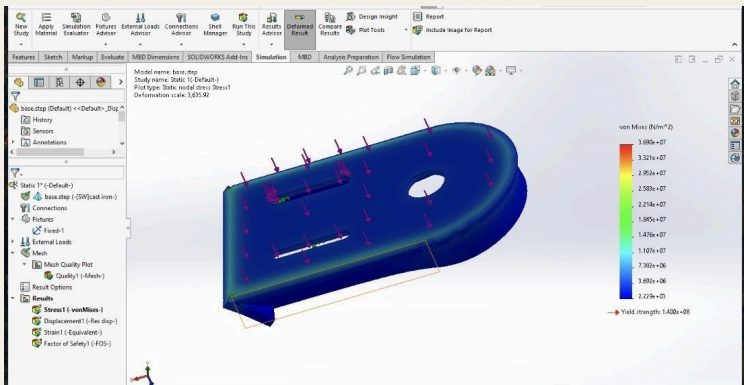
First natural frequency verified at 24.84 Hz.

04 Material Testing

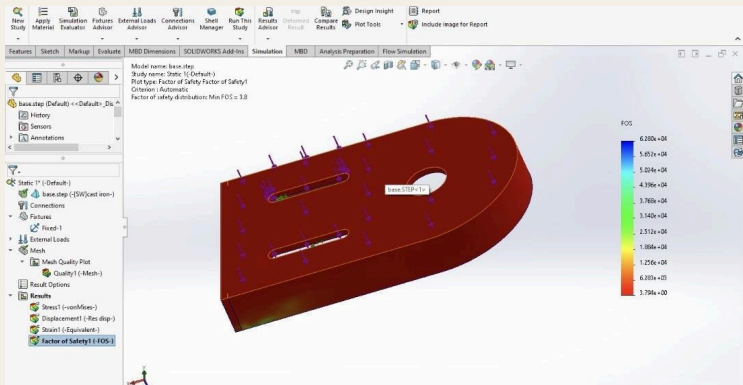
Compression and hardness testing of deposited AA6061.

Base Plate - Structural & Vibration Analysis

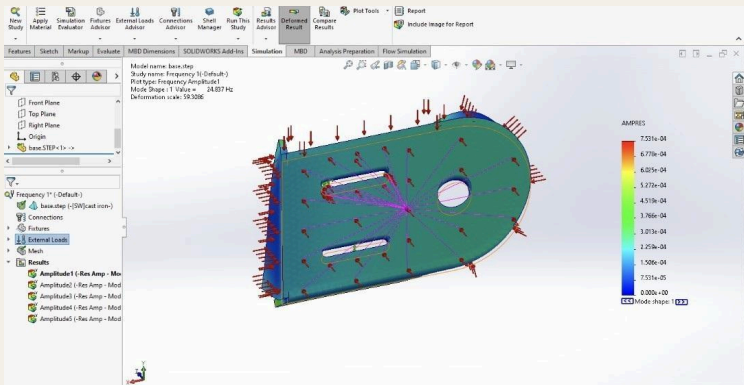
Static structural and modal analysis verified operating-load capacity and frequency separation before fabrication.



VON MISES STRESS



FACTOR OF SAFETY



MODAL FREQUENCY

OBJECTIVE

Verify the base under operating loads and locate critical stress.

METHOD

Static structural study plus first-mode frequency evaluation.

RESULT

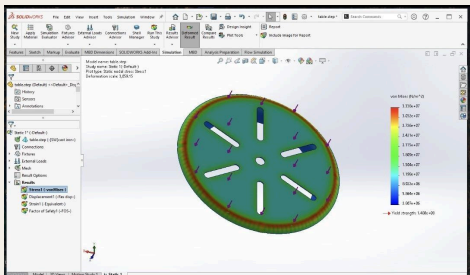
36.9 MPa stress vs. 140 MPa yield; FOS 3.8; first mode 24.84 Hz.

Table & Support Column - Structural Analysis

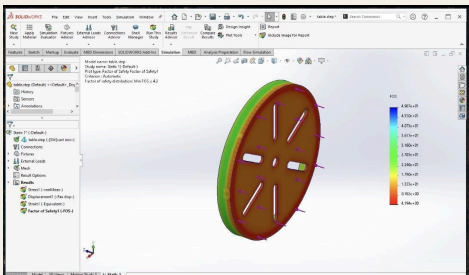
The rotating table plate and vertical support column were verified as critical load-path components of the AFSD machine.

02 / SIMULATION PROJECT

TABLE PLATE



STRESS



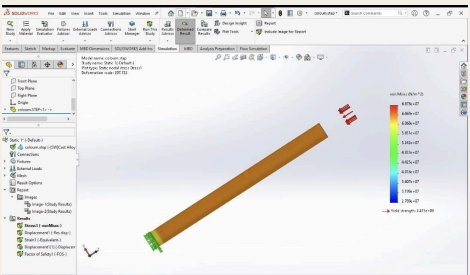
SAFETY FACTOR

Verified Result

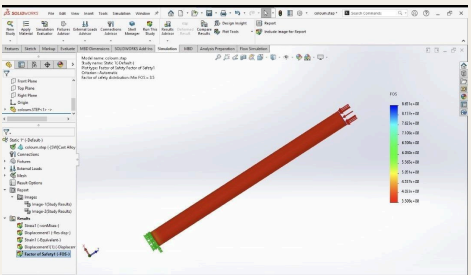
33.4 MPa maximum stress vs. 140 MPa yield. Validated FOS: 4.2.

03 / SIMULATION PROJECT

SUPPORT COLUMN



STRESS



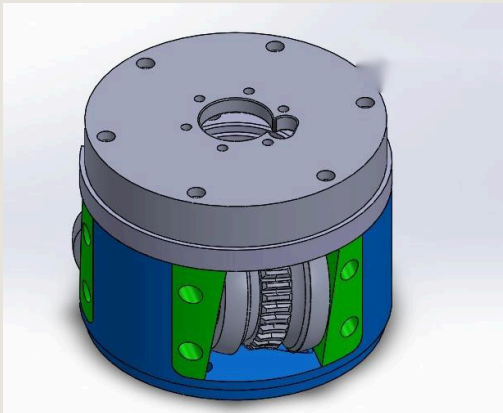
SAFETY FACTOR

Verified Result

68.8 MPa maximum stress vs. 241 MPa yield. Validated FOS: 3.5.

Advanced Systems Design at Tech Valley Islamabad

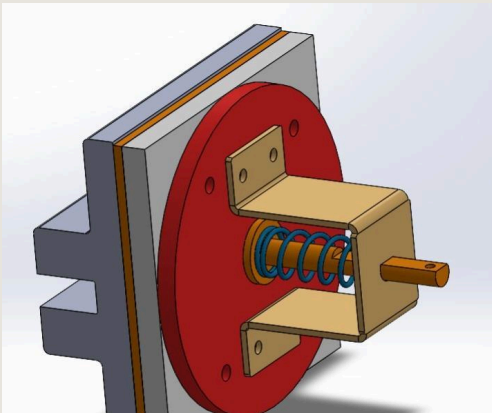
Mechanical assemblies for drone handling, rotary cleaning equipment, and vehicle-mounted structural systems.



01 / SolidWorks

Rotary Cylindrical Cleaner - Assembly Design

PURPOSE	Rotary cleaning of internal cylindrical bore surfaces.
DESIGN WORK	Designed the cylindrical housing, internal drive gear train, and top mounting flange with bolt hole pattern.
PROJECT CONTEXT	Design Engineer Tech Valley, Islamabad



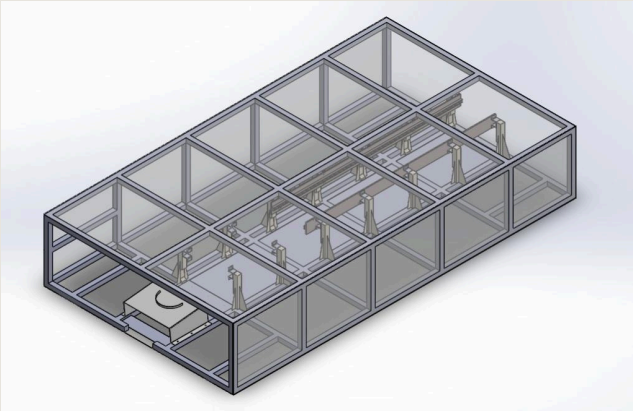
02 / SolidWorks

Drone Locking System - Mechanism Design

PURPOSE	Secure a drone payload in position.
DESIGN WORK	Designed the locking arms, guide pin, compression spring assembly, and mounting flange.
PROJECT CONTEXT	Design Engineer Tech Valley, Islamabad

Advanced Systems Design at Tech Valley Islamabad

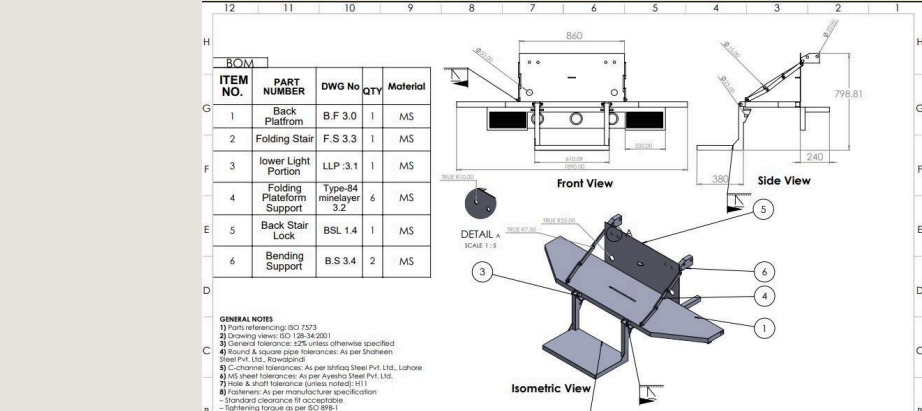
Mechanical assemblies for drone handling, rotary cleaning equipment, and vehicle-mounted structural systems.



03 / SolidWorks

Drone Deployment Rack - Multi-Cell System

PURPOSE	Sequential deployment from a modular multi-cell frame.
DESIGN WORK	Designed the deployment station grid and sliding drive mechanism for sequential drone release.
PROJECT CONTEXT	Design Engineer Tech Valley, Islamabad



04 / SolidWorks

Folding Stair & Platform Mechanism - Vehicle-Mounted Assembly

PURPOSE	Vehicle mounted folding access system.
DESIGN WORK	Prepared the mechanism, supports, locks, views, BOM, tolerance notes, and fabrication details.
PROJECT CONTEXT	Design Engineer Tech Valley, Islamabad

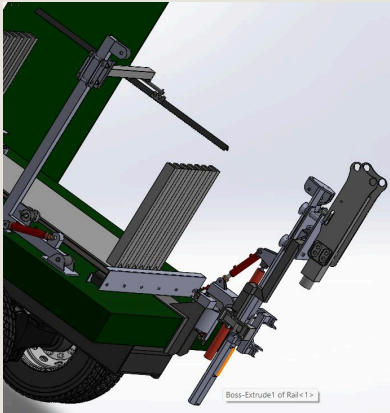
Advanced Systems Design at Tech Valley Islamabad

Vehicle-mounted mechanized system for automated perimeter fence installation.

05 / SolidWorks

Perimeter Fencing Rig - Vehicle-Mounted Assembly

PURPOSE	Mechanized fence-laying system replacing manual perimeter marking.
DESIGN WORK	Designed the full vehicle-mounted assembly integrating storage racks, fence-roll feed reels, and the rear picket driving rail.
PROJECT CONTEXT	Design Engineer Tech Valley, Islamabad



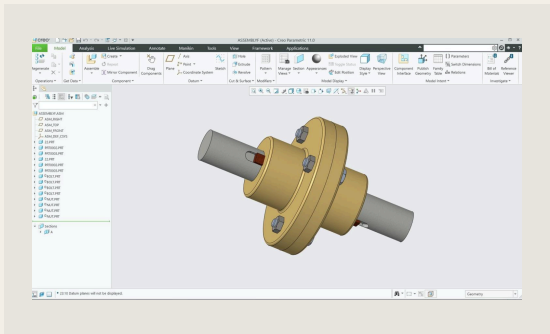
06 / SolidWorks

Automatic Hammer & Rail Mechanism - Fence-Post Driver

PURPOSE	Automated picket-driving unit that replaces manual hammering in the field.
DESIGN WORK	Designed the hydraulic hammer, guide rail, and actuator-pin linkage that drive fence posts into the ground.
PROJECT CONTEXT	Design Engineer Tech Valley, Islamabad

PTC Creo Design & CNC Programming

Internship work covering part and assembly modelling, CNC laser cutting and manufacturing setup, toolpath generation, and NC code.

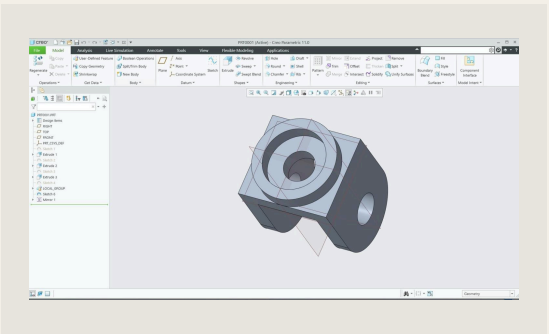


ASSEMBLY DESIGN

Flange Coupling Assembly Design with Fastener Pattern

Bolt-hole fastener pattern and shaft interface modelled in PTC Creo.

01 / NUST Internship

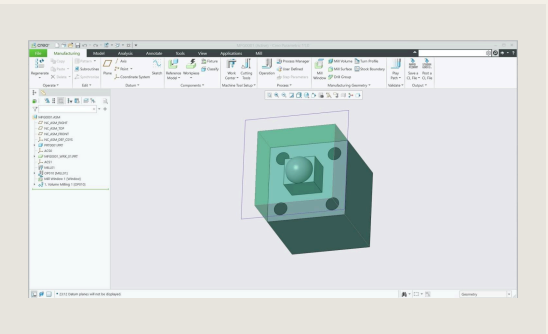


PART DESIGN

Profile-Milled Component Part Design

Stepped bores and a chamfered profile prepared for CNC machining.

02 / NUST Internship

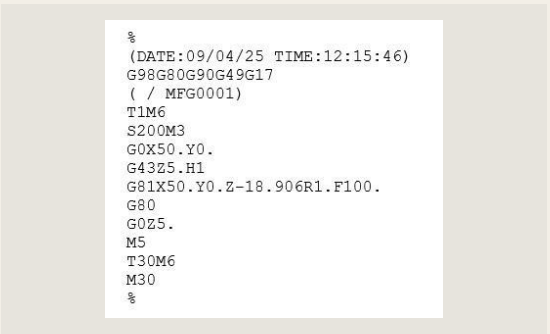


MANUFACTURING

CNC Manufacturing Setup Volume Milling Operation

Volume-milling setup and toolpath generation in PTC Creo Manufacturing.

03 / NUST Internship



CNC PROGRAMMING

Generated NC Code (G-code / M-code)

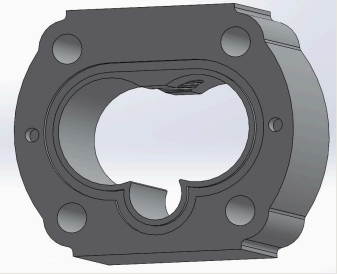
NC program generated from the PTC Creo manufacturing operation.

04 / NUST Internship

01 Model	02 Toolpath	03 G/M Code	04 Machine
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Component & Assembly Design Work

SolidWorks work completed during the NUST internship and the AFSD final-year project.

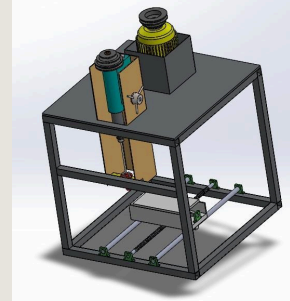


SOLIDWORKS COMPONENT DESIGN

Housing Design

Head housing with motor mount interface, and bolt-hole pattern.

NUST College of EME | Internship

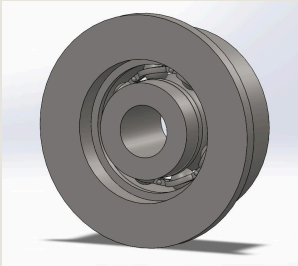


SOLIDWORKS ASSEMBLY DESIGN

AFSD Machine Frame & Table Assembly Design

Complete structural frame and worktable with motor mounting, linear guides, and base support.

HITEC University Taxila | Final Year Project

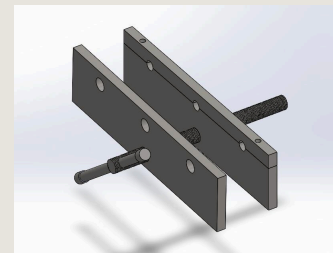


SOLIDWORKS POWER TRANSMISSION

Bearing & Pulley Housing Design

Bearing and pulley housing assembly sized to standard bearing tolerances.

NUST College of EME | Internship



SOLIDWORKS FIXTURE DESIGN

Bolted Clamp Plate Fixture

Bolted clamp-plate fixture with an adjustable bolt hole pattern for workpiece alignment.

NUST College of EME | Internship

Component & Assembly Design Work

University and internship projects are identified separately with their verified project context.

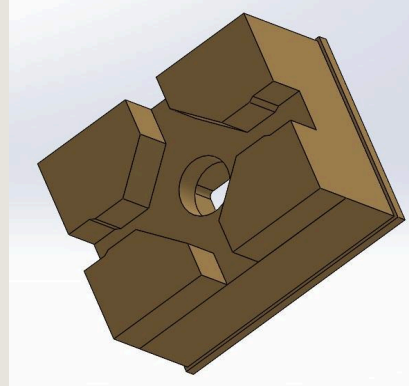


SOLIDWORKS UNIVERSITY PROJECT

Vertical Milling Machine Head

Vertical milling machine head and related components modelled for FYP understanding.

HITEC University Taxila | Final Year Project



SOLIDWORKS MACHINED COMPONENT

Machined Bracket Component

Mounting bracket with stepped faces and a central bore, prepared for VMC machining.

NUST College of EME | Internship



SOLIDWORKS SEMESTER PROJECT

Tail Stock of Lathe Machine

Manual hand tail stock assembly modelled for lathe machine operations.

HITEC University Taxila | Semester Project

AUTOCAD PROJECTS

AutoCAD Design Projects

Residential design work completed while learning AutoCAD through the NAVTTC six-month diploma course.

COURSE PROJECT

Residential 3D House Plan and Layout

Completed industry-based architectural design work in AutoCAD, including a residential 3D house plan and layout.

SOFTWARE	AutoCAD
COURSE	NAVTTC 6-month AutoCAD Diploma
PROJECT TYPE	Residential 3D house plan and layout



Additional Academic Projects

Academic projects on SolidWorks and AutoCAD.

01 Heat Transfer

Fins Design Verification under Forced Convection

Verified circular and rectangular fin designs by comparing actual lengths with optimized theoretical lengths for maximum heat-transfer performance.

HITEC University Taxila

02 Renewable Energy System

Wind-Driven Water Pump

Designed a wind-driven mechanical system for water pumping in agricultural areas using renewable energy instead of conventional power.

HITEC University Taxila

03 Vibration Analysis

Power Transmission Mechanism for Vibration Analysis

Designed and developed a mechanical power-transmission mechanism for demonstration in vibration-analysis experiments.

HITEC University Taxila

AVAILABLE FOR MECHANICAL DESIGN OPPORTUNITIES

Let's turn a complex requirement
into a *working design*.

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NAQEEB ARSHAD

MECHANICAL DESIGN ENGINEER / RAWALPINDI, PAKISTAN